Lesson 6: Artemis and the Future of Space Exploration

NASA and Space Exploration: A Journey through the Solar System

Image credit: NASA

Recap from Last Week

- We learned all about space shuttles!
- How many space shuttles were there?
  - 5
  - Columbia, Challenger, Discovery, Atlantis, Endeavour
- We learned about the parts of a space shuttle.
- What are these parts?
  - Engine, Cockpit, loading bay, solid rocket booster, tank, wings, heat shield

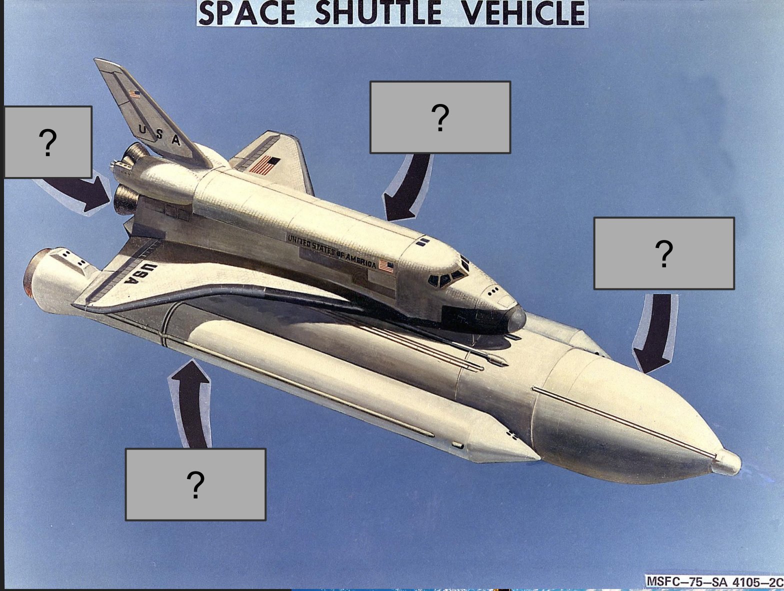
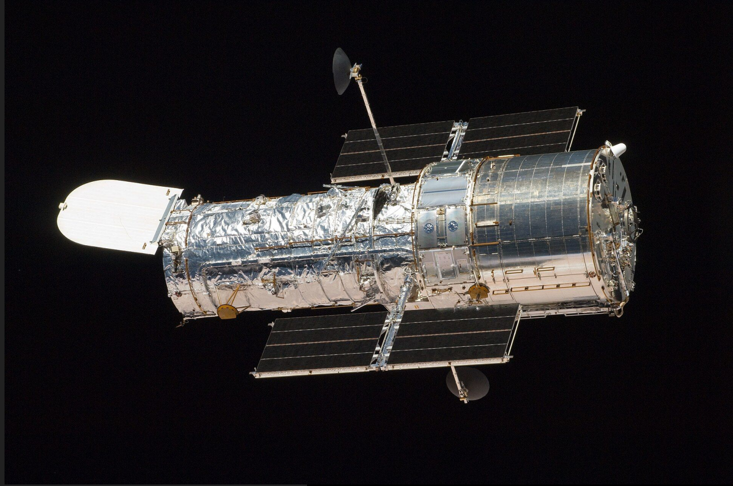


Image credit: NASA

Recap from Last Week

- The space shuttles have done 135 Missions.
- What did the astronauts do during these flights?
  - Deployed new satellites into space
  - Repaired the Hubble Space Telescope
  - Flew to the ISS
  - 8 classified missions for the Military (???)
- What was wrong with the Hubble Space Telescope?
  - The mirror was to flat

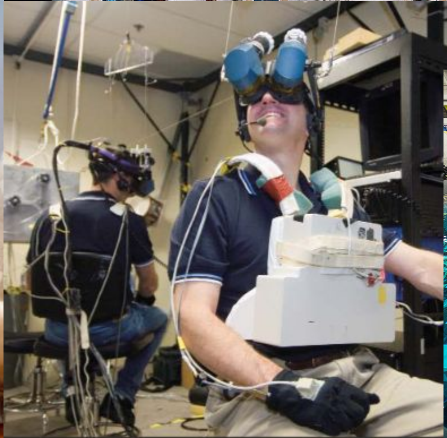


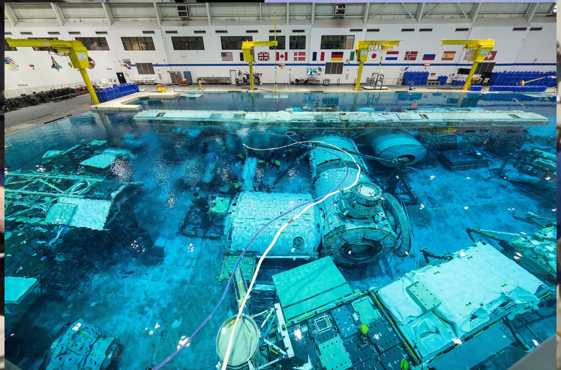
Your favourite HST / JWST image!

Matthew's Favourite!

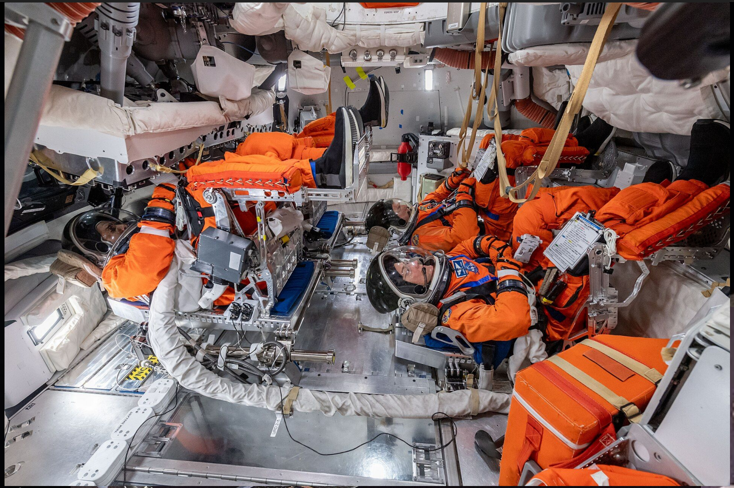
Sven's Favorite!

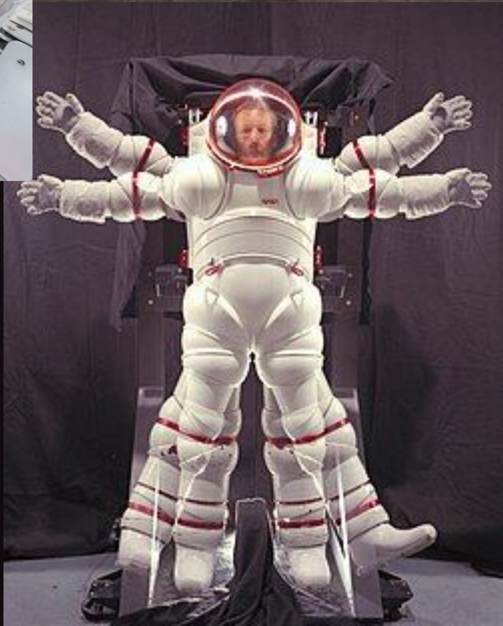
Fall Classes: Astronaut Training





Today’s Topic: Artemis and the Future of NASA





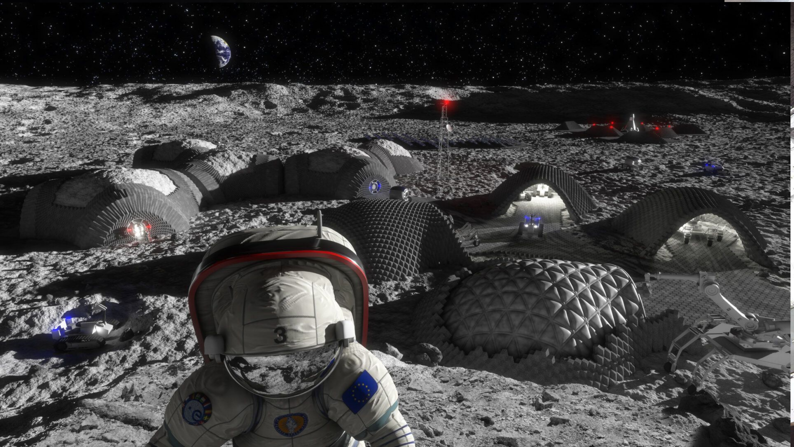
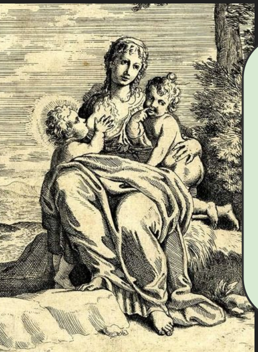


Image credit: NASA

The Artemis Mission

- There are 5 Artemis missions
  - 1: Send a satellite to the Moon (2022)
  - 2: Send humans around the Moon and back (happened this year!)
  - 3: Test docking maneuvers (2027)
  - 4: Send humans onto the Moon (early 2028)
  - 5: Start building a moon base (late 2028)



Who is the **Artemis** programm named after?

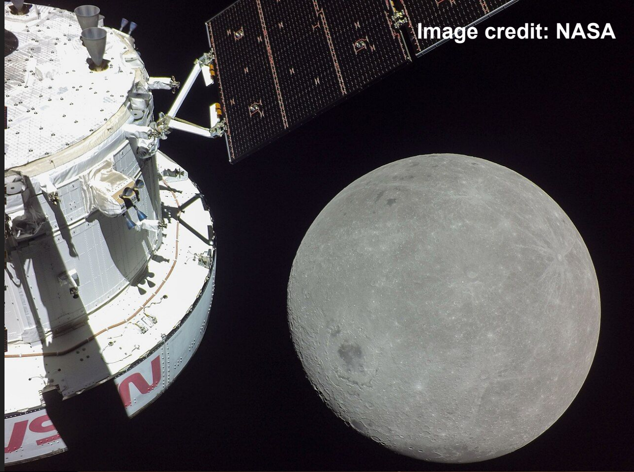
The goddess of hunting, the wilderness, wild animals, transitions, nature, vegetation

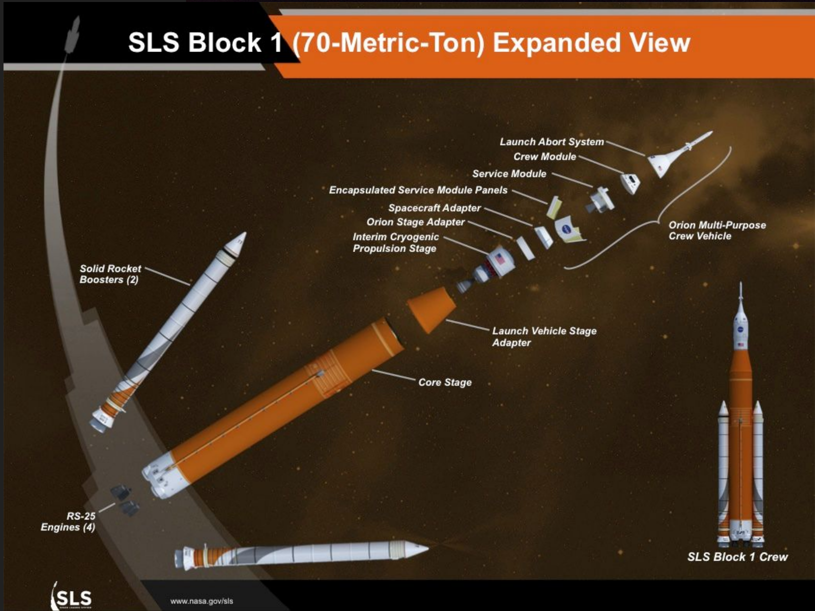
Why is this space program called Artemis?

Artemis is the twin sister of Apollo!

Artemis I

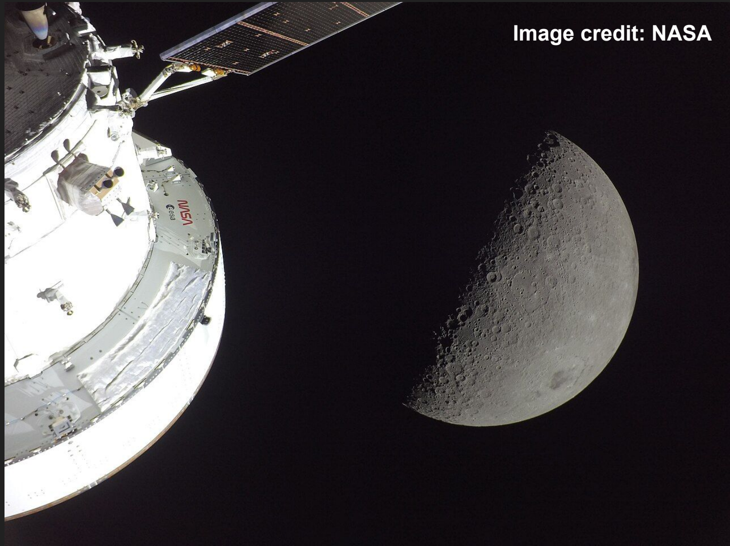
- Artemis I traveled to the Moon and back
- There were no humans on board.
- The goal was to test if the spacecraft is ready for humans
- The Space Launch System (SLS) is similar to the Saturn V rocket and has multiple stages

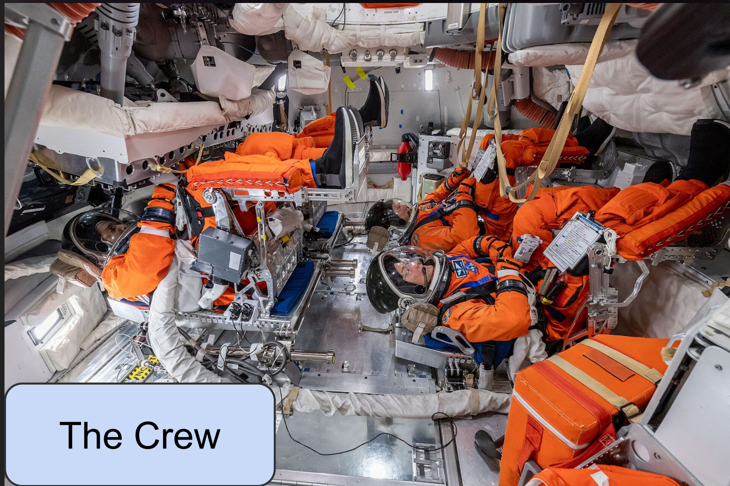




Artemis II

- Artemis II traveled to the Moon and back
- They did not land on the Moon
- The goal was to test the spaceship, the crew capsule and the landing
- They started on April 1st 2026
  - Does anyone remember how long it takes to get to the Moon and back?
  - It takes 4 days!
  - The crew landed April 11th
- Everything went according to plan!





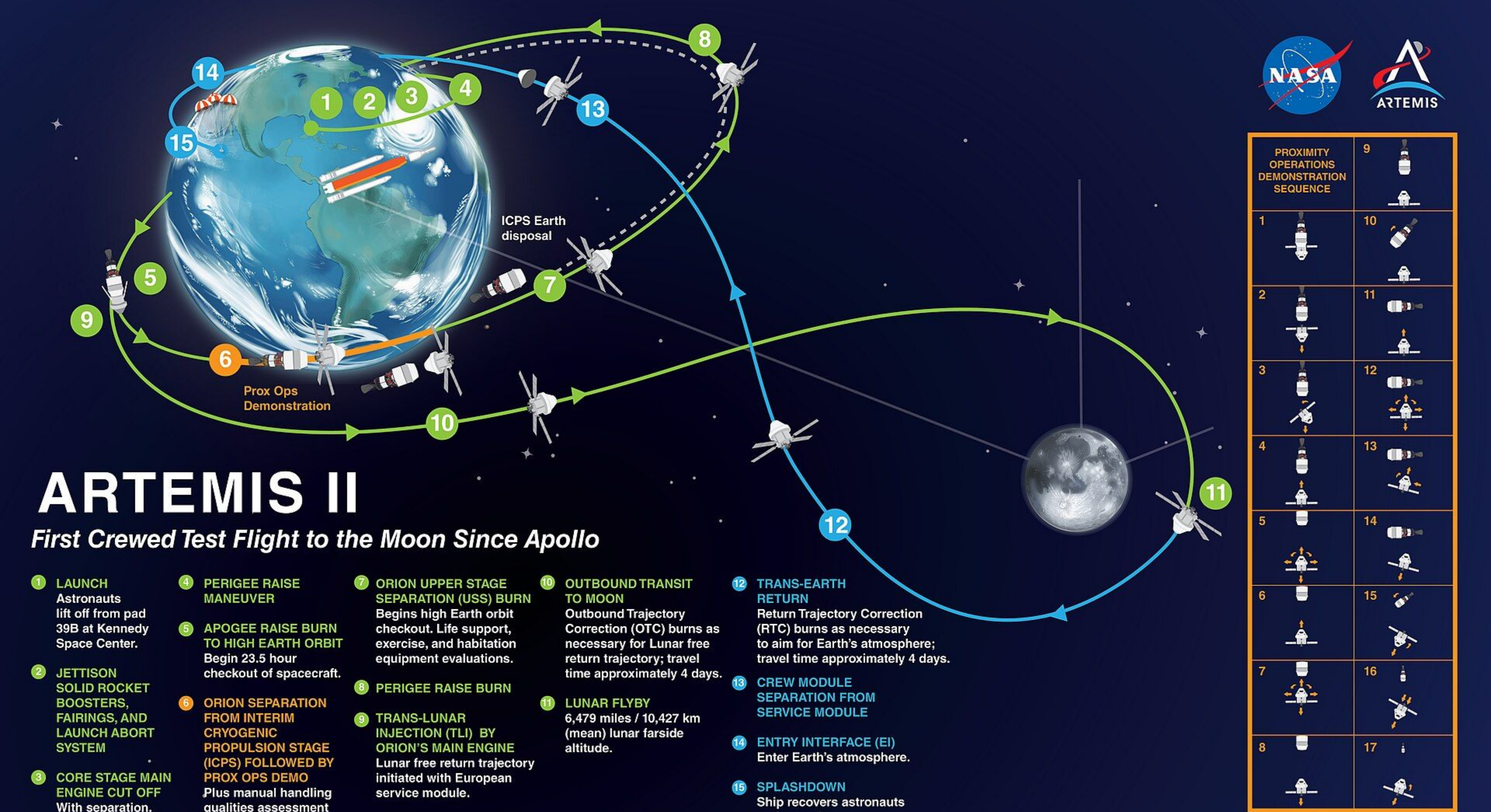


Image credit: NASA

Artemis II

Artemis II

The Artemis II

- The crew saw:
  - The Moon!
  - Earth vanishing behind the moon (An Earthset :D)
  - A solar eclipse!

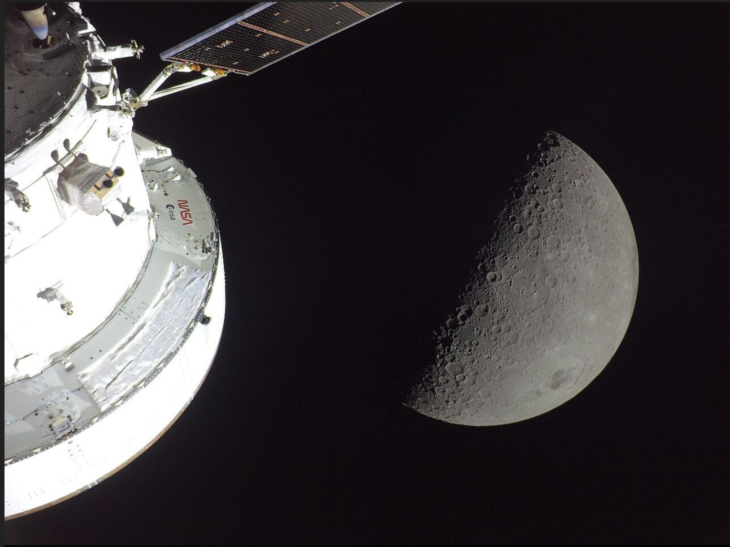


Image credit: NASA

Artemis II

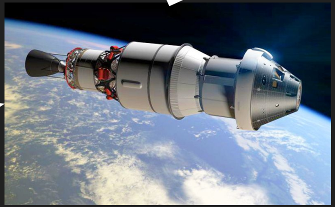
Image credit: NASA / SpaceX / Blue Origins

Artemis III

- Artemis III prepares for the docking of multiple systems

Step 1: Blue Origins Module is launched

Step 2: Artemis III is launched with crew



Step 3: Artemis docks with Blue Origins

Step 5: Artemis crew tests the module

Step 6: Artemis leaves Blue origin, flies to SpaceX and docks

Step 7: SpaceX rocket is launched

Step 8: Artemis Crew tests the module

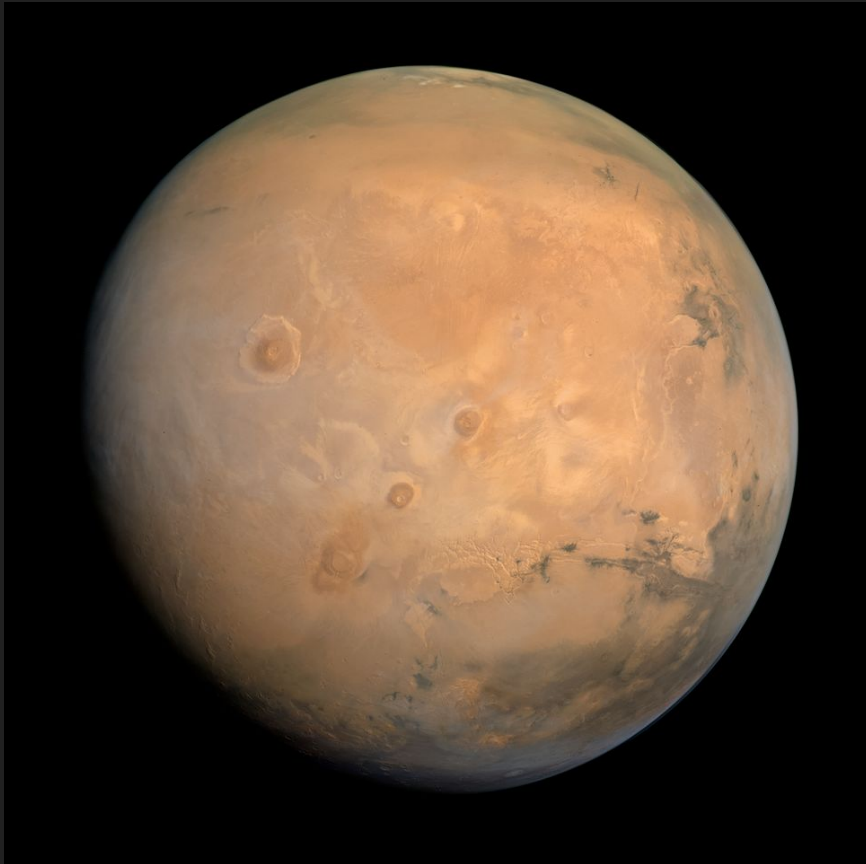
Step 9: Artemis and its crew returns alone to Earth

The Artemis IV and V

- These missions will bring humans back onto the Moon
- Little is currently known about them
- Artemis IV will be a landing + surface vehicle (car) to explore the moon
- Artemis V might be the first step in building a Moon base
    - It is unclear if this will happen

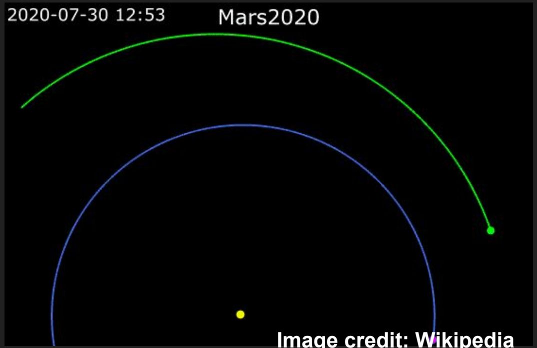
Big Next Step: Living on Mars

- Many missions to colonize Mars has been proposed.
- Currently, no space agency is seriously planning to go to Mars and establish a base there.
- Why is that?

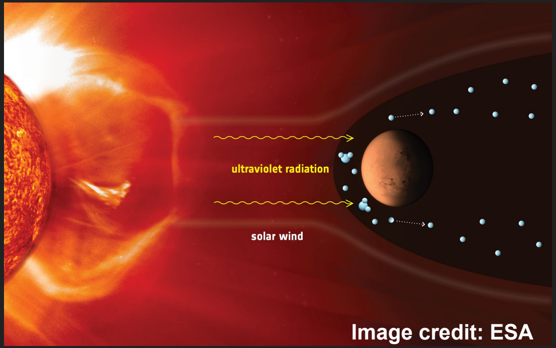


Living on Mars - Challenges

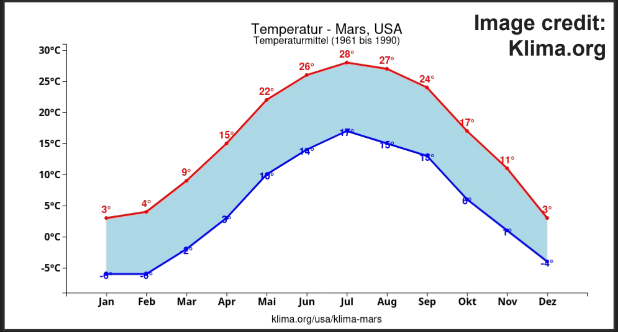
Distance from Earth



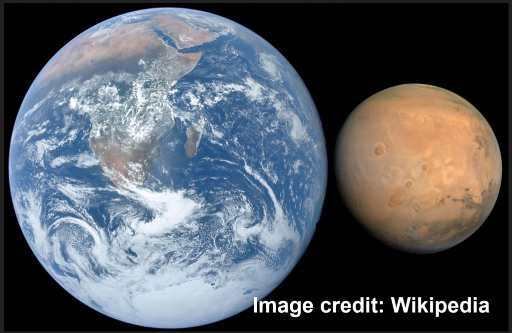
Radiation from the Sun



Extreme Temperatures



Low Gravity



Living on Mars - Distance from Earth

- Problem: It takes a long time to reach Mars
- Solutions:
  - Travel when Mars and Earth are close
  - Spaceships were human can survive for a long time
- Shortest travel time is
  - But this window exists only every 15 years

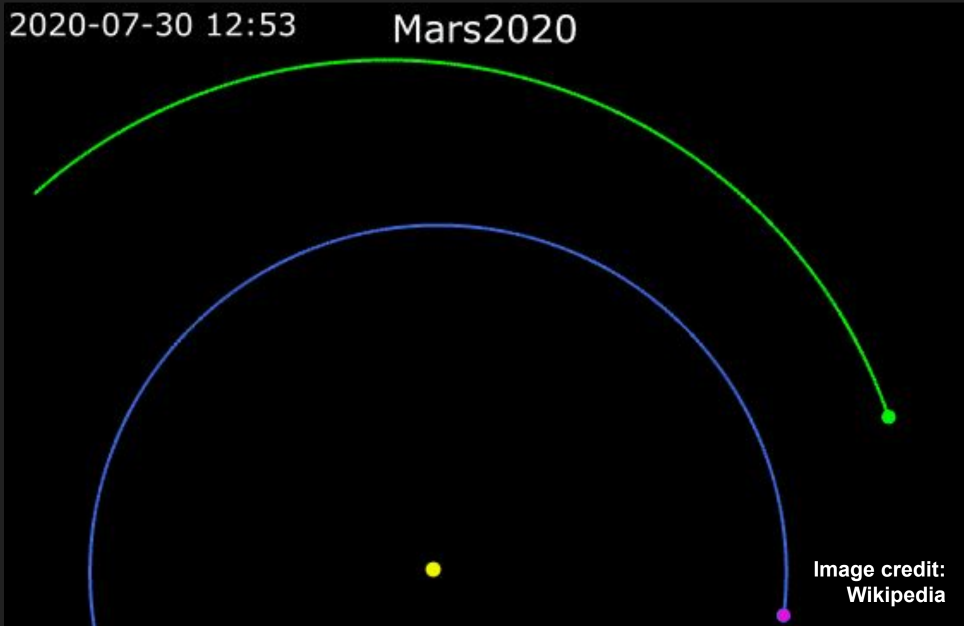


Image credit: Wikipedia

Living on Mars - Radiation

- Problem: Mars has a thin atmosphere and radiation is dangerous
- Solutions:
    - Underground space houses
    - Scifi: Magnetic shielding
    - Scifi: Changing the planets Atmosphere (Terraforming)

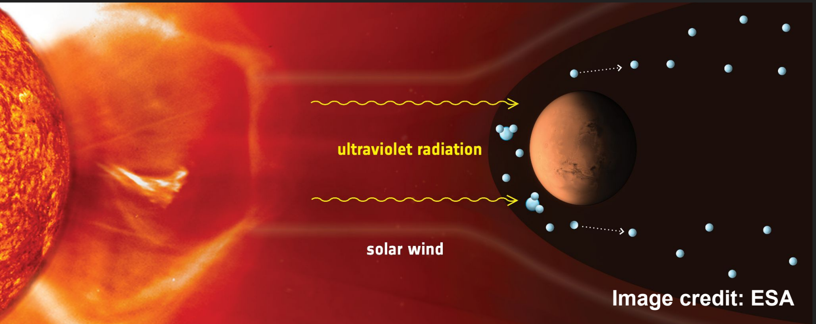


Image credit: ESA

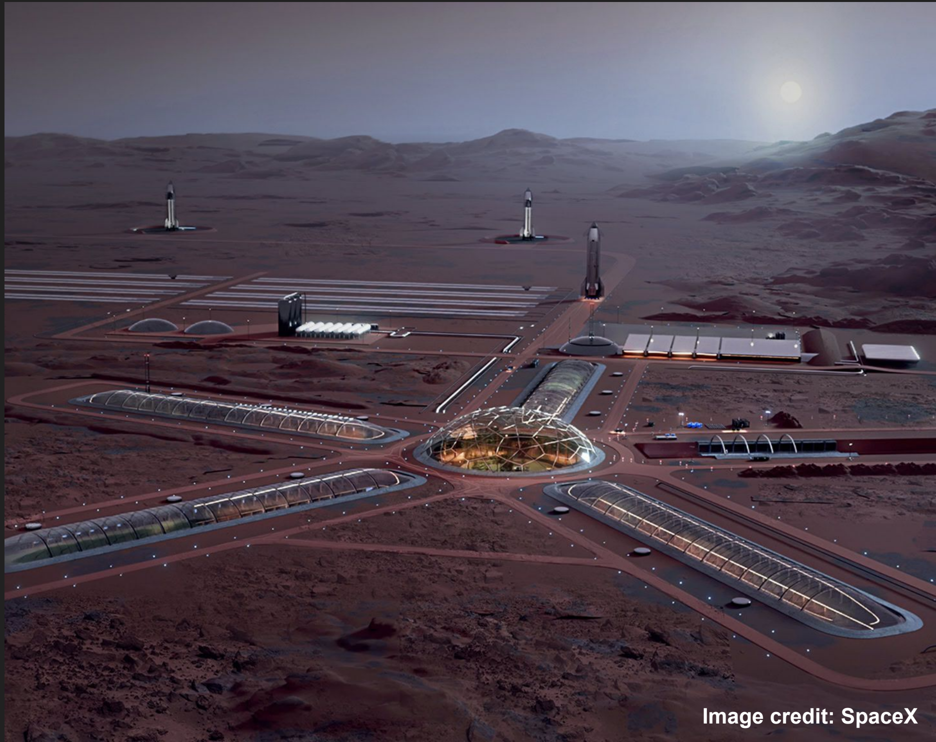
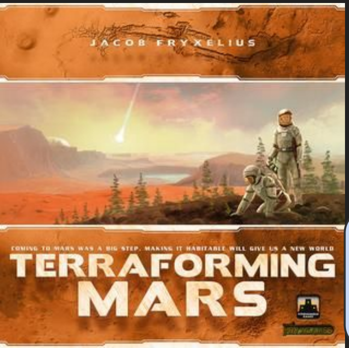


Image credit: SpaceX

Living on Mars - Extreme Temperatures

- Problem: Mars is too cold for humans to survive
- Solutions:
  - Heated houses
  - Heated space suits
  - Scifi: Changing the planets Atmosphere (Terraforming)



There is a board game about terraforming Mars!

Image credit: NASA

Living on Mars - Low Gravity

- Problem: The low gravity of Mars is not healthy for humans
- Solutions:
    - Going to the gym!
    - Scifi: Over hundreds of years, humans will adapt to the low gravity

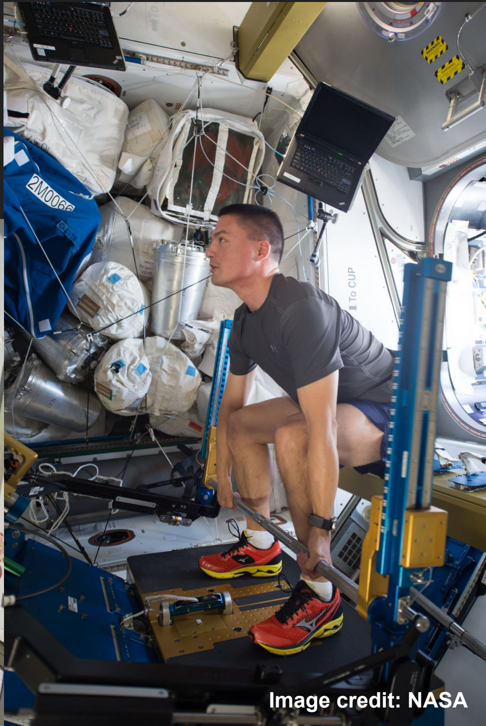


Image credit: NASA

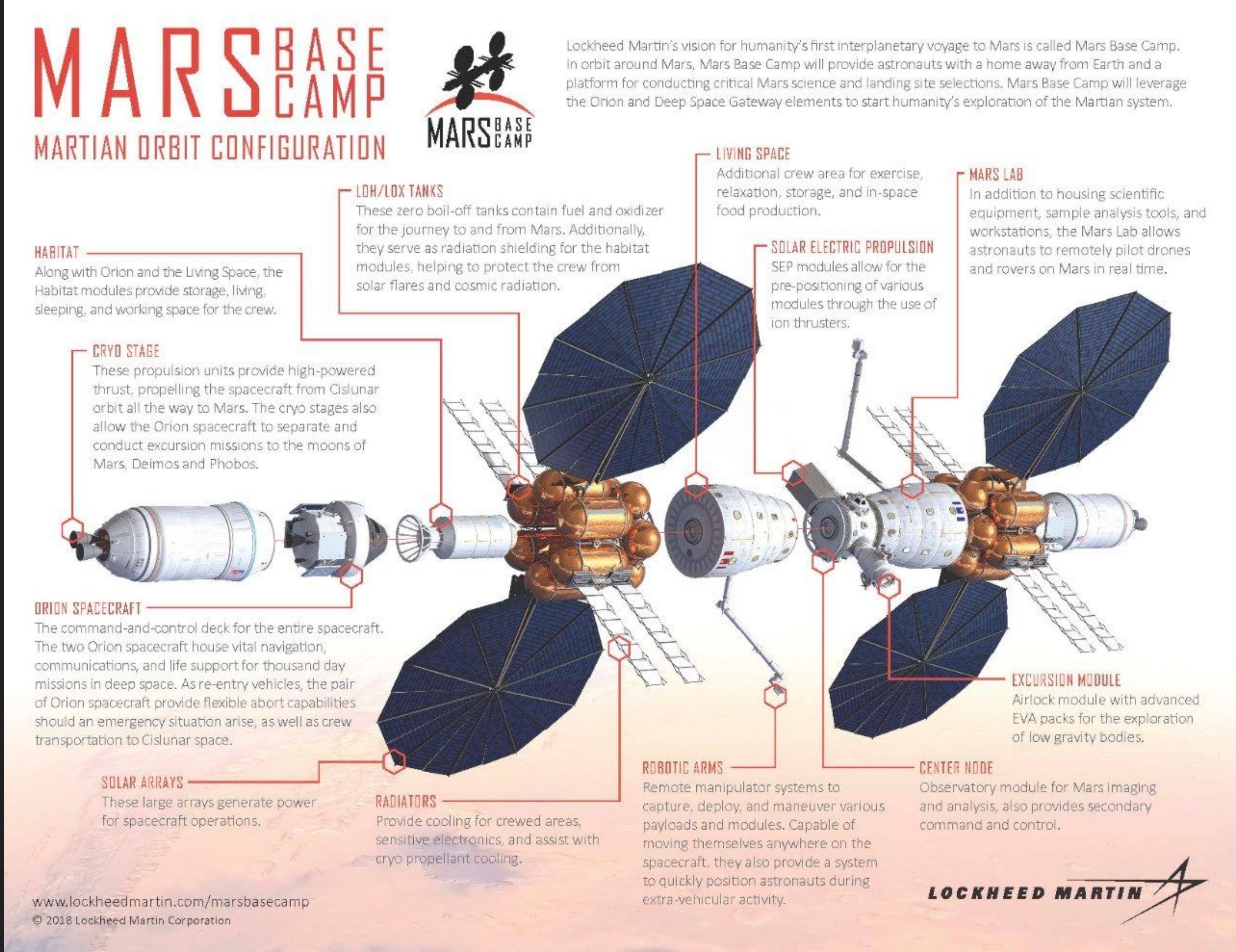

MARS
TECH

Future NASA missions

- NASA is launching new Telescopes like Nancy Grace Roman and SunRISE (several small solar telescopes)
- Collaborations with Japan Aerospace Exploration Agency and Roscosmos to supply the ISS
- Work together with Blue Origin, Intuitive Machines and Firefly to deliver technology to the lunar south pole
- And of course, Artemis III, IV, and V!

Why are we not covering Future missions in detail?

- Flying into space is hard
- Missions tend to be delayed:
    - Artemis II: 2019 ➪ 2023 ➪ 2025 ➪ 2026
    - JWST: 2007 ➪ 2013 ➪ 2018 ➪ 2020 ➪ 2021
- Missions tend to change what they do until they launch:
    - Artemis III: Originally planned to be the first human landing on the Moon
    - Artemis V: The Lunar Gateway Orbital Station is postponed
- While delays are annoying, they are an important part of the process!
- Space flight is expensive and there are no shortcuts.
- **Safety must always comes first!**


MARS BASE CAMP
MARTIAN ORBIT CONFIGURATION
MARS BASE CAMP
Lockheed Martin's vision for humanity's first interplanetary voyage to Mars is called Mars Base Camp. In orbit around Mars, Mars Base Camp will provide astronauts with a home away from Earth and a platform for conducting critical Mars science and landing site selections. Mars Base Camp will leverage the Orion and Deep Space Gateway elements to start humanity's exploration of the Martian system.
HABITAT
Along with Orion and the Living Space, the Habitat modules provide storage, living, sleeping, and working space for the crew.
LOH/LOX TANKS
These zero boil-off tanks contain fuel and oxidizer for the journey to and from Mars. Additionally, they serve as radiation shielding for the habitat modules, helping to protect the crew from solar flares and cosmic radiation.
LIVING SPACE
Additional crew area for exercise, relaxation, storage, and in-space food production.
SOLAR ELECTRIC PROPULSION
SEP modules allow for the pre-positioning of various modules through the use of ion thrusters.
MARS LAB
In addition to housing scientific equipment, sample analysis tools, and workstations, the Mars Lab allows astronauts to remotely pilot drones and rovers on Mars in real time.
CRYO STAGE
These propulsion units provide high-powered thrust, propelling the spacecraft from Cislunar orbit all the way to Mars. The cryo stages also allow the Orion spacecraft to separate and conduct excursion missions to the moons of Mars, Deimos and Phobos.
ORION SPACECRAFT
The command-and-control deck for the entire spacecraft. The two Orion spacecraft house vital navigation, communications, and life support for thousand day missions in deep space. As re-entry vehicles, the pair of Orion spacecraft provide flexible abort capabilities should an emergency situation arise, as well as crew transportation to Cislunar space.
EXCURSION MODULE
Airlock module with advanced EVA packs for the exploration of low gravity bodies.
SOLAR ARRAYS
These large arrays generate power for spacecraft operations.
RADIATORS
Provide cooling for crewed areas, sensitive electronics, and assist with cryo propellant cooling.
ROBOTIC ARMS
Remote manipulator systems to capture, deploy, and maneuver various payloads and modules. Capable of moving themselves anywhere on the spacecraft, they also provide a system to quickly position astronauts during extra-vehicular activity.
CENTER NODE
Observatory module for Mars imaging and analysis, also provides secondary command and control.
www.lockheedmartin.com/marsbasecamp
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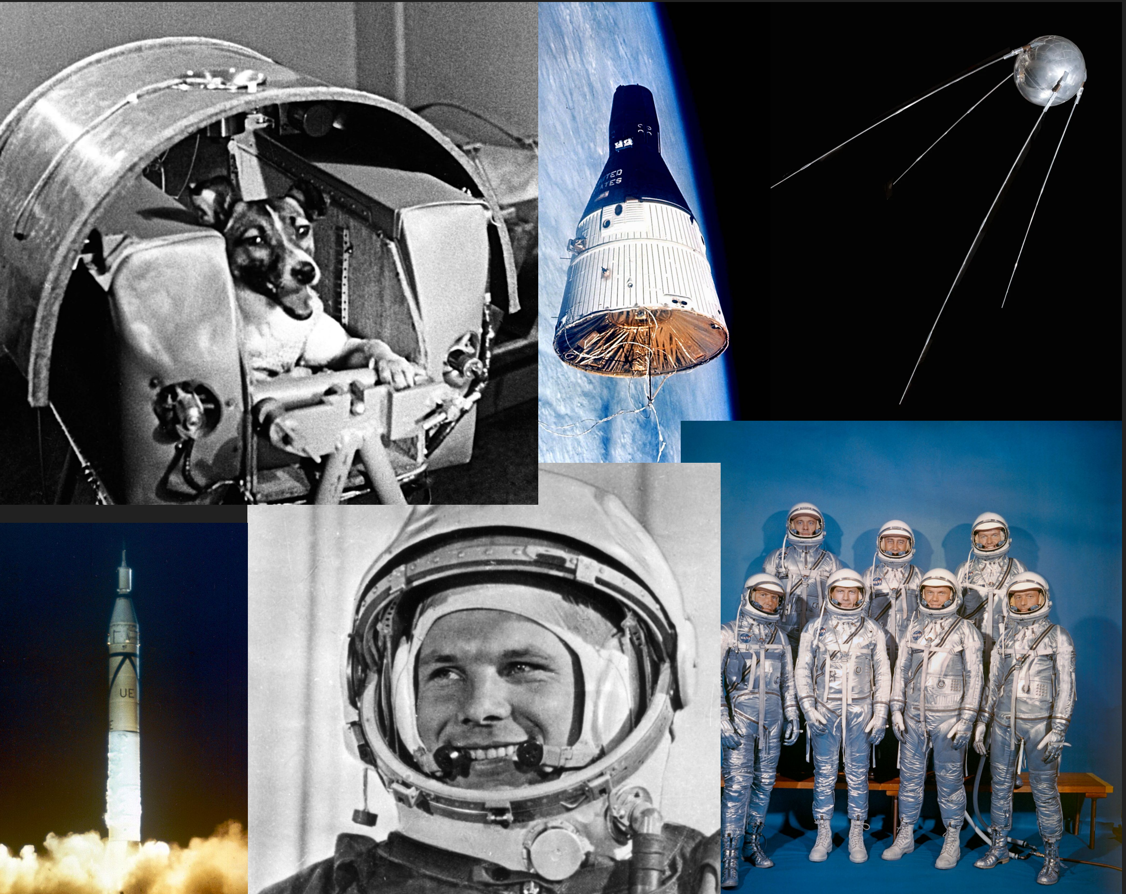
Today’s Activity: Design your Mars colony!

- Take a sheet of paper
- Draw different buildings and connect them
- Think about where the people sleep, eat, where the food comes from, where energy comes from etc.

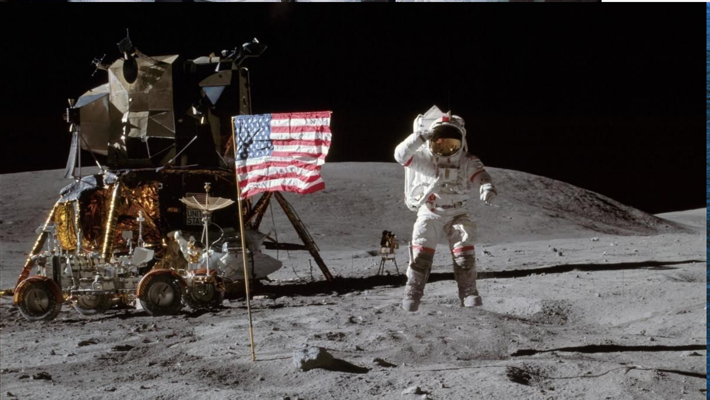
Lecture 1: Earth Orbit and the Space Race



- First animal in space Laika
- First satellite: Sputnik
- First Human in space
- NASA missions: Project Mercury and Project Gemini
- We learned how rockets work and launched air rockets

What's the name of the first human in space?
Yuri Gagarin

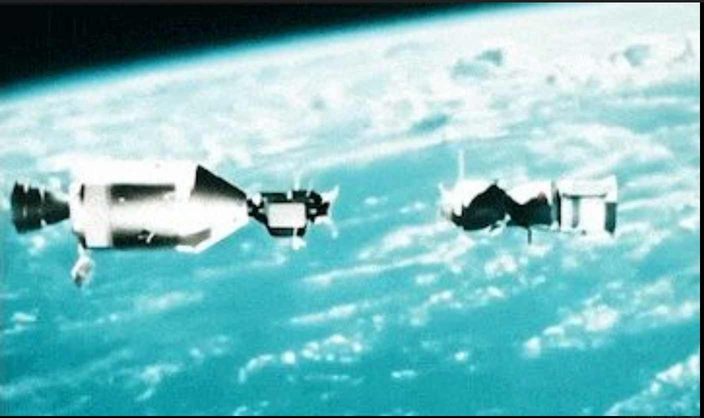
Lecture 2: The Apollo program

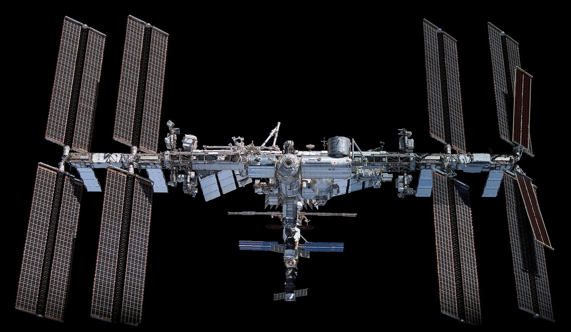


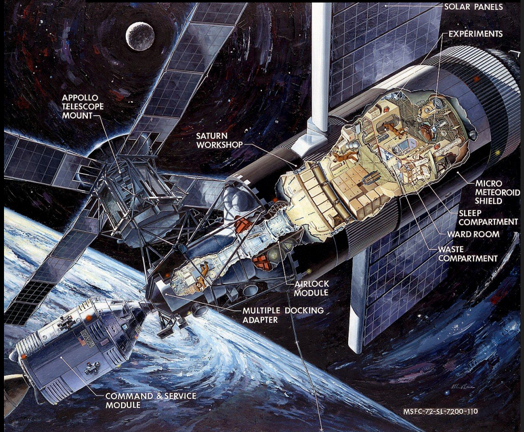
- There were 17 Apollo missions
- Apollo 11 brought the first humans to the Moon
- We learned the different stages of the Saturn V rocket
- Apollo 13 had a problem ….
- We built our own egg-landers!

Who was the first human that stepped onto the Moon?
Neil Armstrong

Lecture 3: Technology Orbiting Earth



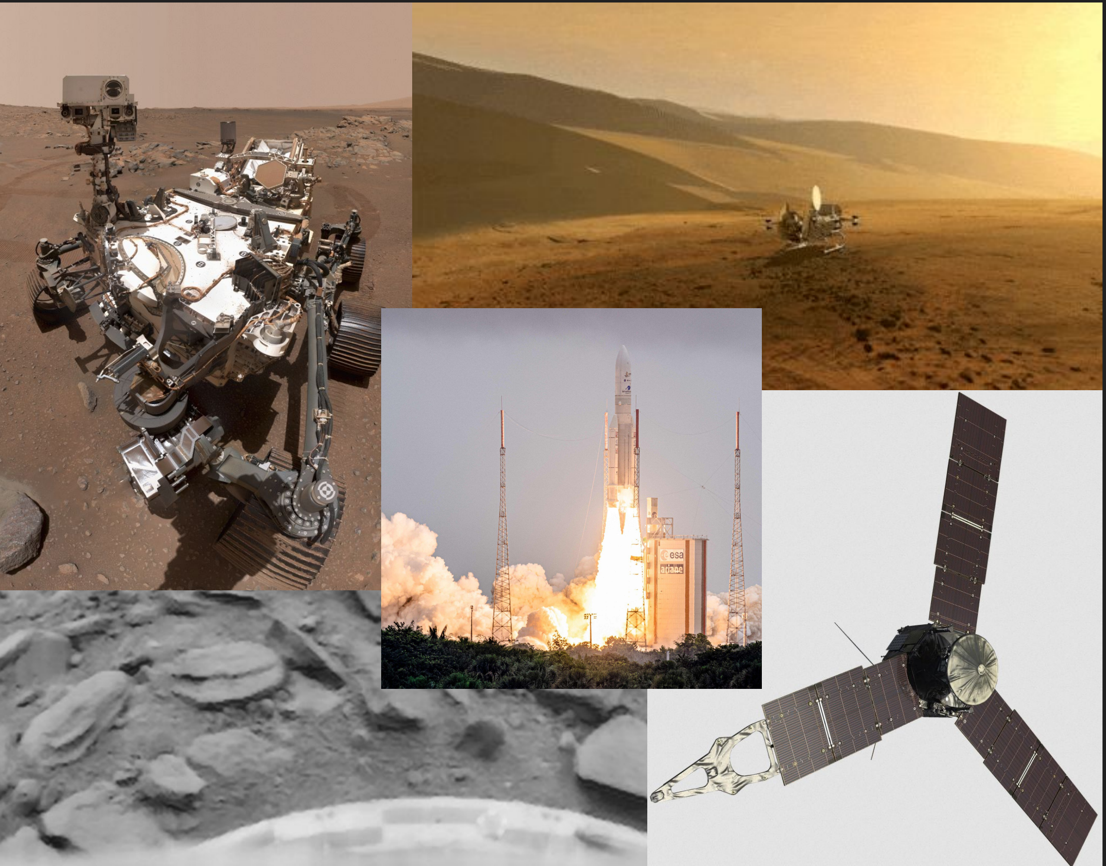




- Apollo-Soyuz Handshake
- The first space stations: Salyut, Skylab, and MIR
- How astronauts live in space, how they use the toilet and how they sleep

What is the name of the most famous space station?
International Space Station (ISS)

Lecture 4: Rovers and Probes



- Mars, Venus, Jupiter and the Sun have been visited often
- Saturn, Neptune, Uranus, and Pluto have barely been visited
- We learned all the parts of satellites and mars rovers
- We tested our own little rover!

Which Mars rover had its birthday last week?
Curiosity

Lecture 5: The Space Shuttle Programm

- There were 5 space shuttles: Columbia, Challenger, Discovery, Atlantis, Endeavour
- In total they flew 135 Missions
- They deployed satellites, repaired HST, and flew to the ISS
- We had a paper airplane contest!

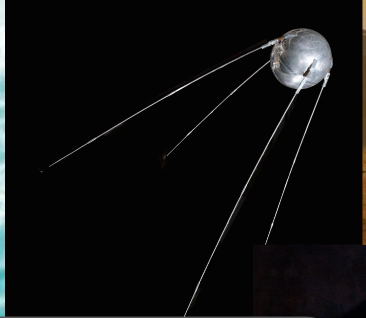
Which space shuttle flew the most missions?
Discovery (39)

Lecture 6: Artemis and NASA's Future

- Artemis II went around the Moon and saw an Earth set and a space-solar eclipse
- Artemis IV will go back to the Moon
- Living on Mars is hard because of the radiation, cold, and low gravity
- Future Moon bases

How long does it take to fly to the Moon?
4 days






Now you know all about NASA and how we explore space!